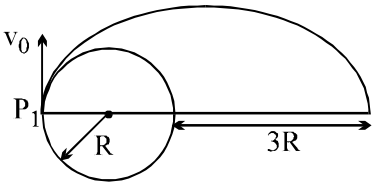
SECTION-I

Single Correct Answer Type

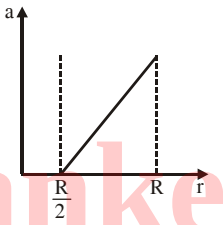
8 Q. [3 M (–1)]

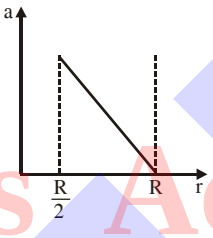
1. The acceleration due to gravity on the surface of earth is g & radius R. A ball is projected horizontally from point P_1 on the surface of earth (as shown in the figure) If the maximum height attained by the ball above the earth surface is 3R then speed of projection V_0 is (neglect cosmic dust resistance & earth rotation about its axis)

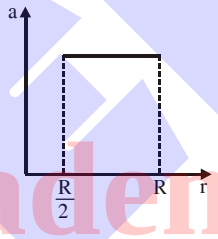


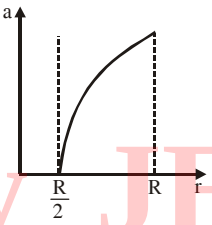
(A) $\sqrt{\frac{8}{5}gR}$ (B) $\sqrt{2gR}$ (C) $\sqrt{gR}$ (D) $\sqrt{\frac{6}{5}gR}$

2 A frictionless tunnel is dug along a chord of the earth at a perpendicular distance $\frac{R}{2}$ from the centre of earth (where R is radius of earth). An object is released from one end of the tunnel. The correct graph, showing the variation of acceleration of particle with its distance r from centre of earth is :

(A) 

(B) 

(C) 

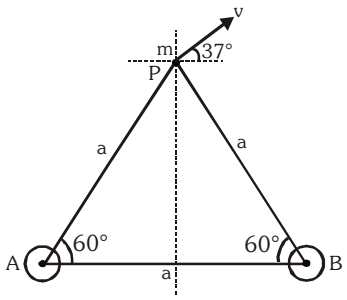
(D) 

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3. A satellite of mass m is orbiting a planet of mass M at a radial distance r_0 from the centre of the planet. The satellite expels a small mass in a direction opposite to its orbital velocity. The immediate recoil velocity of the satellite exceeds the initial orbital velocity by Δv. The largest possible value of Δv for which the satellite remains within the gravitational field of the planet is

(A) $\sqrt{\frac{2GM}{r_0}}$ (B) $\sqrt{\frac{GM}{2r_0}}$ (C) $\left(\sqrt{2}-1\right)\sqrt{\frac{GM}{r_0}}$ (D) None of these

4. With what minimum speed should m be projected from point P in presence of two fixed spherical masses M each at A and B is shown in figure such that mass m should escape the gravitational attraction of A and B.



(A) $\frac{6}{5}\sqrt{\frac{GM}{a}}$ (B) $\frac{8}{5}\sqrt{\frac{GM}{a}}$

(C) $\sqrt{\frac{2GM}{a}}$ (D) $2\sqrt{\frac{GM}{a}}$

5. A small satellite of mass 'm' is revolving around earth in a circular orbit of radius r_0 with speed v_0. At certain point of its orbit, the direction of motion of satellite is suddenly changed by angle $\theta = \cos^{-1}(3/5)$ by turning its velocity vector in the same plane of motion, such that speed remains constant. The satellite, consequently goes to elliptical orbit around earth. The ratio of speed at perigee to speed at apogee is :-

(A) 3 (B) 9 (C) 1/3 (D) 1/9

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6. If a satellite orbits as close to the earth's surface as possible, then which of the following statement is incorrect
(A) its speed is maximum
(B) time period of its rotation is minimum
(C) the total energy of the earth plus satellite system minimum
(D) the total energy of the earth plus satellite system is maximum

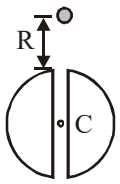
7. Satellites A and B are orbiting around the earth in orbits of ratio R and 4R respectively. The ratio of their areal velocities is :
(A) 1 : 2 (B) 1 : 4 (C) 1 : 8 (D) 1 : 16

8. Four particles, each of mass M and equidistant from each other, move along a circle of radius R under the action of their mutual gravitational attraction. The speed of each particle is :
(A) $\sqrt{\frac{GM}{R}(1+2\sqrt{2})}$ (B) $\frac{1}{2}\sqrt{\frac{GM}{R}(1+2\sqrt{2})}$ (C) $\sqrt{\frac{GM}{R}}$ (D) $\sqrt{2\sqrt{2}\frac{GM}{R}}$

Multiple Correct Answer Type 7 Q. [4 M (–1)]

9. Two point objects of masses m and 4m are at rest at an infinite separation. They move towards each other under mutual gravitational attraction. If G is the universal gravitational constant, then at a separation r
(A) the total mechanical energy of the two objects is zero
(B) their relative velocity is $\sqrt{\frac{10Gm}{r}}$
(C) the total kinetic energy of the objects is $\frac{4Gm^2}{r}$
(D) their relative velocity is zero

10. A particle is dropped from a height equal to the radius of the earth above the tunnel dug through the earth as shown in the figure.
R : Radius of earth ; M : Mass of earth
(A) Particle will oscillate through the earth to a height R on both sides
(B) Particle will execute simple harmonic motion
(C) Motion of the particle is periodic
(D) Particle passes the centre of earth with a speed = $\sqrt{\frac{2GM}{R}}$



11. Two spherical bodies of masses M and 2M and radii R and 2R, respectively, start approaching each other at time t = 0 from rest, due to mutual gravitational attraction. They were initially very far away from each other. They collide at time t = T.
(A) Distance travelled by their COM till time t = T is zero (COM–centre of mass)
(B) Their relative velocity at time t = T is $\frac{1}{3}\sqrt{\frac{2GM}{R}}$
(C) PE of the system decreases as the bodies approach each other
(D) In COM frame, speeds of the bodies are always equal

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12. A particle at a distance r from the centre of a uniform spherical planet of mass M radius R (< r) has a velocity of magnitude v.

(A) for $0 < v < \sqrt{\frac{GM}{r}}$ trajectory may be ellipse

(B) for $v = \sqrt{\frac{GM}{r}}$ trajectory may be ellipse

(C) for $\sqrt{\frac{GM}{r}} < v < \sqrt{\frac{2GM}{r}}$ trajectory may be ellipse.

(D) for $v = \sqrt{\frac{GM}{r}}$ trajectory may be circle

13. Two tunnels are dug from one side of the earth's surface to the other side, first along a diameter and the second along a chord. Now two particles are dropped from one end of each of the tunnels. Both the particles oscillate simple harmonically along the tunnels. Let T_1 and T_2 be the time period v_1 and v_2 be the maximum speed of the particle in the two tunnels. Then

(A) $T_1 = T_2$ (B) $T_1 > T_2$ (C) $v_1 = v_2$ (D) $v_1 > v_2$

14. Satellite A is in a circular orbit of radius r and satellite B in another circular orbit of radius 4 r, both revolving around the earth. The masses of the satellites A and B are in the ratio 3 : 1. If the symbols V, E, u and T represent the speed, total energy, escape velocity and period of revolution of satellite, then

(A) $V_A/V_B=2$ (B) $E_A/E_B=12$ (C) $u_A/u_B=2$ (D) $T_A/T_B=\frac{1}{8}$.

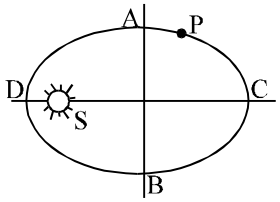
15. A partical of mass m is thrown in a parabolic path towards earth having mass M and radius R. It comes to a greatest approach distance R/2 from surface of earth. Magnitude of total energy change required to make the particle moving in an elliptical path having eccentricity 1/2, with the greatest approach point either aphelion or perihelion are :

(A) $\frac{1}{3}\frac{GMm}{R}$ (B) $\frac{1}{6}\frac{GMm}{R}$ (C) $\frac{1}{2}\frac{GMm}{R}$ (D) $\frac{GMm}{R}$

Linked Comprehension Type (1 Para × 2Q.) [3 M (-1)]
(Single Correct Answer Type)

Paragraph for Question No. 16 and 17

Figure shows the orbit of a planet P around the sun S. AB and CD are the minor and major axes of the ellipse.



16. If t_1 is the time taken by the planet to travel along ACB and t_2 the time along BDA, then

(A) $t_1 = t_2$ (B) $t_1 > t_2$
(C) $t_1 < t_2$ (D) nothing can be concluded

17. If U is the potential energy and K kinetic energy then $|U| > |K|$ at

(A) Only D (B) Only C (C) both D & C (D) neither D nor C

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Matching List Type (4 × 4) 1Q.[3 M (–1)]

18. A planet of mass M, has two natural satellites with masses m_1 and m_2. The radii of their circular orbits are R_1 and R_2 respectively. Ignore the gravitational force between the satellites. Define v_1, L_1, K_1 and T_1 to be, respectively, the orbital speed, angular momentum, kinetic energy and time period of revolution of satellite 1 ; and v_2, L_2, K_2 and T_2 to be the corresponding quantities of satellite 2. Given $m_1/m_2 = 2$ and $R_1/R_2 = 1/4$, match the ratios in List-I to the numbers in List-II.

	List–I		**List–II**
P.	$\frac{v_1}{v_2}$	1.	$\frac{1}{8}$
Q.	$\frac{L_1}{L_2}$	2.	1
R.	$\frac{K_1}{K_2}$	3.	2
S.	$\frac{T_1}{T_2}$	4.	8

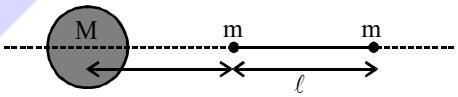
(A) P → 4 ; Q → 2 ; R → 1 ; S → 3 (B) P → 3 ; Q → 2 ; R → 4 ; S → 1
(C) P → 2 ; Q → 3 ; R → 1 ; S → 4 (D) P → 2 ; Q → 3 ; R → 4 ; S → 1

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) 2 Q. [4 M (0)]

19. A binary star consists of two stars A (mass 2.2 M_S) and B (mass 11 M_S), where M_S is the mass of the sun. They are separated by distance d and are rotating about their centre of mass, which is stationary. The ratio of the total angular momentum of the binary star to the angular momentum of star B about the centre of mass is:-

20. A large spherical mass M is fixed at one position and two identical point masses m are kept on a line passing through the centre of M (see figure). The point masses are connected by a rigid massless rod of length ℓ and this assembly is free to move along the line connecting them. All three masses interact only through their mutual gravitational interaction. When the point mass nearer to M is at a distance $r = 3\ell$ from M, the tension in the rod is zero for $m = k\left(\frac{M}{288}\right)$. The value of k is :



SECTION-IV

Matrix Match Type (4 × 5) 1 Q. [8 M (for each entry +2(0)]

21. In elliptical orbit of a planet, as the planet moves from apogee position to perigee position,

	Column–I		**Column–II**
(A)	Speed of planet	(P)	Remains same
(B)	Distance of planet from centre of Sun	(Q)	Decreases
(C)	Potential energy	(R)	Increases
(D)	Angular momentum about centre of Sun	(S)	Can not say

Subjective Type 5 Q. [4 M (0)]

22. A particle of mass m was transferred from the centre of the base of a uniform hemisphere of mass M and radius R into infinity. What work was performed in the process by the gravitational force exerted on the particle by the hemisphere?

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23. A pair of stars rotates about a common center of mass. One of the stars has a mass M which is twice as large as the mass m of the other. Their centres are at a distance d apart, d being large compared to the size of either star. (a) Derive an expression for the period of rotation of the stars about their common centre of mass in terms of d,m, G. (b) Compare the angular momentum of the two stars about their common centre of mass by calculating the ratio L_m/L_M. (c) Compare the kinetic energies of the two stars by calculating the ratio K_m/K_M.

24. Given a thin homogeneous disc of radius a and mass m_1. A particle of mass m_2 is placed at a distance l from the disc on its axis of symmetry. Initially both are motionless in free space but they ultimately collide because of gravitational attraction. Assuming a << l, find the relative velocity at the time of collision.

25. A small body of mass m is projected with a velocity just sufficient to make it reach from the surface of a planet (of radius 2R and mass 3M) to the surface of another planet (of radius R and mass M). The distance between the centers of the two spherical planets is 6R. The distance of the body from the center of bigger planet is 'x' at any moment. During the journey, find the distance x where the speed of the body is (a) maximum (b) minimum. Assume motion of body along the line joining centres of planets.

26. A remote sensing satellite is revolving in an orbit of radius x over the equator of earth. Find the area on earth surface in which satellite can not send message.

GR # GRAVITATION **ANSWER KEY**

SECTION-I

Single Correct Answer Type **8 Q. [3 M (–1)]**

1. Ans. (A) **2 Ans. (D)** **3. Ans. (C)** **4. Ans. (D)** **5. Ans. (B)**
6. Ans. (D) **7. Ans. (A)** **8. Ans. (B)**

Multiple Correct Answer Type **7 Q. [4 M (–1)]**

9. Ans. (A, B, C) **10. Ans. (A,C, D)** **11. Ans. (A,C)** **12. Ans. (A,B,C,D)** **13. Ans. (A, D)**
14. Ans. (A,B,C,D) **15. Ans. (B,C)**

Linked Comprehension Type **(1 Para × 2Q.)** **[3 M (-1)]**
(Single Correct Answer Type)

16. Ans. (B) **17. Ans. (C)**

Matching List Type (4 × 4) **1Q.[3 M (–1)]**

18. Ans. (B)

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **2 Q. [4 M (0)]**

19. Ans. 6 **20. Ans. 7**

SECTION-IV

Matrix Match Type (4 × 5) **1 Q. [8 M (for each entry +2(0)]**

21. Ans. (A)-R, (B)-Q, (C)-Q (D)-P

Subjective Type **5 Q. [4 M (0)]**

22. Ans. $A = -\frac{3}{2}\frac{GmM}{R}$

23. Ans. (a) $T=\frac{2\pi d^{3/2}}{\sqrt{3Gm}}$, (b) 2, (c) 2

24. Ans. $\left[2G(m_1+m_2)\left(\frac{2}{a}-\frac{1}{l}\right)\right]^{1/2}$

25. Ans. (a) 2R, (b) $3\sqrt{3}\left(\sqrt{3}-1\right)R$

26. Ans. $\left(1-\frac{\sqrt{x^2-R^2}}{x}\right)4\pi R^2$

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